

Swaraaha: A Comparative Study of Deep Learning Architectures for Speech Dysfluency Classification and Localization

K. SHREEKRISHNA UPADHYAYA, M. CHETHAN KESHAV BHAT, SKANDA PRASAD K, AND SRINIVAS HEGDE
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*Department of Artificial Intelligence & Machine Learning,
Vivekananda College of Engineering & Technology, Puttur, Karnataka, India*

{shreekrishna, chethan, skanda, srinivas}@vcet.ac.in

Abstract — Stuttering is a prevalent speech disorder that manifests as involuntary repetitions, prolongations, and blocks, significantly impacting communication and quality of life. Automated detection and temporal localization of dysfluency events remain challenging due to the subtle acoustic signatures, severe class imbalance, and the need for fine-grained temporal precision. This paper presents **Swaraaha**, an end-to-end speech dysfluency analysis system, and provides a comprehensive comparative study of seven architectural variants spanning three model families: Wav2Vec 2.0-based classifiers (five independent binary classifiers and a shared-backbone multitask variant), and convolutional neural network (CNN) spectrogram models (single-head, pooled, LSTM-augmented, and transformer-augmented). All models are evaluated on a merged corpus of 37,087 clips drawn from SEP-28K, UCLASS, and Project Boli, using both in-distribution held-out and cross-corpus evaluation protocols. Our results demonstrate that the shared-backbone Wav2Vec 2.0 multitask classifier achieves the highest in-distribution macro F1 (0.5215), closely followed by the five independent binary classifiers (0.5183), while requiring $\approx 4.9\times$ fewer parameters than five separate models. CNN-based models show strong cross-corpus generalization on Boli, with the CNN-LSTM architecture achieving 0.5206 F1 — substantially outperforming Wav2Vec2 models on unseen data. We analyze per-class performance across all five dysfluency types, discuss the trade-offs between representation capacity and generalization, and provide actionable insights for building robust clinical-grade stutter detection systems.

Keywords — *Speech dysfluency detection, stuttering classification, Wav2Vec 2.0, multitask learning, temporal localization, deep learning, speech signal processing*

I. INTRODUCTION

Stuttering affects approximately 1% of the global adult population and up to 5% of children, making it one of the most common communication disorders [1]. Characterized by involuntary repetitions of sounds, syllables, or words (sound repetitions and word repetitions), abnormal prolongations of speech sounds (prolongations), and involuntary pauses or stops in speech flow (blocks), stuttering significantly impacts the psychosocial well-being of affected individuals [2]. Traditional assessment by speech-language pathologists (SLPs) relies on subjective auditory perception and visual inspection of waveforms — a process that is time-consuming, prone to inter-rater variability, and unsuitable for large populations [3].

Recent advances in self-supervised speech representation learning, particularly Wav2Vec 2.0 [4], have demonstrated remarkable performance on downstream speech classification tasks by learning powerful contextualized representations from raw audio. However, the application of such models to stuttering detection presents unique challenges: the five dysfluency types have vastly different acoustic signatures, severe class imbalance exists (most speech segments are fluent), and the precise temporal localization of dysfluency events requires fine-grained frame-level prediction [5].

This paper makes the following contributions:

1. We present **Swaraaha**, a modular, open-source system for speech dysfluency classification and temporal localization, supporting five dysfluency types: prolongation, block, sound repetition, word repetition, and interjection.
2. We conduct a systematic comparative study of seven architectural variants across three model families — five independent Wav2Vec 2.0 binary classifiers, a shared-backbone multitask classifier, and four CNN spectrogram-based models — on a merged multi-corpus dataset.
3. We provide both in-distribution and cross-corpus evaluation, revealing that Wav2Vec2 models excel in-distribution while CNN-LSTM models generalize better to unseen data.
4. We analyze per-class performance across all five dysfluency types, identifying interjection as the most detectable ($F1 = 0.751$) and block as the most challenging ($F1 = 0.383$), consistent with prior literature.

The remainder of this paper is organized as follows: Section II reviews related work. Section III describes the Swaraaha system architecture. Section IV details the experimental setup. Section V presents results and discussion. Section VI concludes with future directions.

II. RELATED WORK

A. Traditional Approaches

Early stutter detection systems relied on handcrafted acoustic features — including jitter, shimmer, spectral flux, and formant trajectories — combined with classical classifiers such as Support Vector Machines (SVMs), Hidden Markov Models (HMMs), and Gaussian Mixture Models (GMMs) [6]. While these methods achieved reasonable accuracy on controlled datasets, they suffered from poor generalization across speakers, accents, and recording conditions due to the limited representational capacity of handcrafted features.

B. Deep Learning Methods

The introduction of end-to-end deep learning approaches eliminated the need for manual feature engineering. Convolutional Neural Networks (CNNs) operating on mel-spectrograms demonstrated early promise for dysfluency detection [3]. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks were subsequently applied to capture temporal dependencies in speech signals [7]. More recently, transformer-based architectures, particularly Wav2Vec 2.0 [4], have achieved state-of-the-art results on a wide range of speech processing tasks by learning contextualized representations from raw audio through self-supervised pre-training on large unlabeled corpora.

C. Wav2Vec2-Based Stutter Detection

Bayerl et al. [1] applied Wav2Vec2 to multi-task stuttering detection on FluencyBank, achieving macro F1 scores in the range of 0.56 – 0.63 across five dysfluency types. Miyahara et al. [5] fine-tuned Wav2Vec2 on the SEP-28K dataset, reporting per-class F1 scores that varied significantly by dysfluency type (interjection: 0.78, prolongation: 0.53, block: 0.30). The Vocametrix model on HuggingFace achieved a weighted-average F1 of 0.67 on SEP-28K-E using Wav2Vec2-Large-XLSR-53. These works established Wav2Vec2 as a strong backbone for stutter detection but focused primarily on classification without systematic architectural comparison.

D. Temporal Localization

While classification answers **what type** of dysfluency is present, clinical applications additionally require knowing **where** in the audio the dysfluency occurs. Temporal localization has received less attention in the stutter detection literature. CNN-based approaches operating on spectrograms have been proposed for event detection [8], but systematic evaluation of localization models on multi-corpus data remains limited. Swaraaha addresses this gap by incorporating both classification and localization pipelines.

E. Multi-Corpus Evaluation

Prior work has largely evaluated on single datasets (either SEP-28K or FluencyBank), making it difficult to assess cross-corpus generalization. The Boli dataset [9] offers a multilingual, multi-accent corpus with word-level annotations, enabling meaningful cross-corpus evaluation. Our work systematically evaluates all seven models on both in-distribution and cross-corpus held-out data.

III. SYSTEM ARCHITECTURE

Swaraaha comprises two independent model pipelines whose outputs are presented jointly: a **classification pipeline** that identifies which dysfluency types are present, and a **localization pipeline** that pinpoints where they occur in the audio. Both pipelines share a common preprocessing module and are accessed through a unified model registry API.

A. Data Pipeline

The system integrates three publicly available stuttering datasets:

1. **SEP-28K** [10]: 32,321 clips annotated for five dysfluency types, with TSV-format labels.
2. **UCLASS** [11]: 4,712 clips in SEP-28K format, sourced from stuttering therapy sessions.
3. **Project Boli** [9]: 54 clips with word-level dysfluency annotations and timestamps, recorded by speakers of five Indian languages (Hindi, Marathi, Telugu, Bengali, and Assamese).

All three datasets are merged into a unified corpus of 37,087 clips with per-clip interval CSVs recording dysfluency events as (`start_sec`, `end_sec`, `type`) tuples. A stratified 80/10/10 train/val/test split is created, yielding 29,296 training clips, 3,662 validation clips, and 3,716 test clips. To guarantee an honest cross-corpus evaluation, all Project Boli clips are pinned to the test split and never appear in the training or validation splits. Audio is resampled to 16 kHz mono, DC-removed, peak-normalized, silence-trimmed, and padded/truncated to a maximum of 3 seconds.

B. Classification Pipeline

1) Five Independent Binary Classifiers:

Each dysfluency type is handled by a separate binary classifier built on facebook/wav2vec2-base (94,569,090 parameters per classifier). The architecture follows the Hugging Face sequence-classification formulation: raw audio $\rightarrow$ Wav2Vec2 encoder $\rightarrow$ attention-mask-weighted mean pooling over time $\rightarrow$ Linear(768 $\rightarrow$ 2) $\rightarrow$ softmax. Each classifier is trained independently with focal loss ($\gamma = 2.0$) to handle class imbalance, AdamW optimizer ($\text{lr} = 3 \times 10^{-5}$, weight decay = 0.01), linear warmup (500 steps), and mixed-precision training on CUDA. Backbone freezing for the first 3 epochs prevents catastrophic forgetting of pretrained representations.

2) Shared-Backbone Multitask Classifier:

A single Wav2Vec2 encoder feeds five parallel per-class heads, each a small MLP — Linear(768 $\rightarrow$ 768) $\rightarrow$ Tanh $\rightarrow$ Linear(768 $\rightarrow$ 2) — operating on a temporal mean pool of the encoder’s hidden states. The total parameter count is 97.3M — only 2.7M more than a single independent classifier, since the backbone is shared. Training uses summed focal loss across all five heads, allowing the shared backbone to learn representations beneficial to all dysfluency types simultaneously.

C. Localization Pipeline

1) CNN Spectrogram Localizer:

A convolutional network operating on 128-bin mel-spectrograms. The architecture uses four convolutional blocks (Conv2d(3 $\times$ 3) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout2d(0.4)) with progressive channel expansion (1 $\rightarrow$ 32 $\rightarrow$ 64 $\rightarrow$ 128). Each block max-pools only along the frequency axis (kernel 2 $\times$ 1), preserving the time resolution throughout; the fourth block collapses the remaining frequency dimension with adaptive average pooling. A per-frame head of two 1 $\times$ 1 convolutions (128 $\rightarrow$ 64 $\rightarrow$ 1) with a sigmoid outputs dysfluency probability per frame — since time is never downsampled, no transposed convolutions are required.

Trained with BCE loss and $\text{pos_weight} = 5.0$ to compensate for the rarity of dysfluent frames (5% of all frames).

2) Wav2Vec2 Temporal Localizer:

Uses the Wav2Vec2 backbone with a temporal attention pooling head: raw audio $\rightarrow$ Wav2Vec2 encoder $\rightarrow$ temporal attention $\rightarrow$ Linear(768 $\rightarrow$ 256) $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(256 $\rightarrow$ 1) $\rightarrow$ sigmoid. Frame resolution is 20 ms (Wav2Vec2 internal subsampling factor = 320 samples at 16 kHz). Backbone freezing for the first 5 epochs, then unfreezing with $10\times$ lower learning rate.

D. Model Registry

All models are loaded through a unified registry API (Classifier, Localizer, Transcriber, ModelRegistry) that decouples model loading from training. The registry reads checkpoint paths and per-class thresholds from a JSON configuration file, enabling seamless model swapping without code changes. Audio preprocessing (resampling, cleaning, normalization, padding) is applied automatically within the API.

IV. EXPERIMENTAL SETUP

A. Model Configurations

We evaluate seven architectural variants, summarized in Table Table 1.

Arm	Architecture	Type	Parameters
arm01	5 $\times$ Wav2Vec2 binary classifiers	classifier	94,569,090
arm02	Wav2Vec2 multitask, freeze 3 epochs	multitask	97,332,362
arm03	Wav2Vec2 multitask, freeze 20 epochs	multitask	97,332,362
arm04	CNN pooling, multitask	multitask	342,030
arm05	CNN single-head	multitask_single	274,950
arm06	CNN + LSTM, multitask	multitask	457,614
arm07	CNN + Transformer, multitask	multitask	540,302

Table 1: Comparative study arms — architectural variants and parameter counts.

B. Training Protocol

All models are trained on a single NVIDIA GPU with the following shared settings: seed = 42, 20 epochs with early stopping (patience = 5), gradient-norm clipping at 1.0, mixed precision in the classification pipelines, batch size = 8 (binary classifiers), 16 (multitask and CNN classifiers), or 4 (Wav2Vec2 localizer), maximum audio length = 3 seconds, sample rate = 16 kHz. Data augmentation includes Gaussian noise injection ($\sigma = 0.005$), time stretching (0.9–1.1 $\times$), pitch shifting (± 1.0 semitones), circular temporal shifting (± 0.1 s), and amplitude scaling (0.8–1.2 $\times$). Preprocessed audio is cached to disk for fast re-runs.

Thresholds for binary classification are tuned on the validation set using Youden’s J statistic ($J = \text{sensitivity} + \text{specificity} - 1$) to find the operating point that maximizes discriminative ability. No test-set or cross-corpus threshold fitting is performed — this ensures honest evaluation.

C. Evaluation Metrics

For classification, we report precision, recall, F1 score, and AUROC per dysfluency type, plus macro-averaged F1 across all five types. For localization, we report frame-level precision, recall, F1, specificity, event-level detection accuracy, mean Intersection over Union (mIoU), and false alarm rate (events per minute).

D. Evaluation Protocol

Two evaluation settings are used:

1. **In-distribution held-out (Test):** 3,715 valid clips from the held-out test split of the merged corpus. Note: due to same-speaker overlap between train and test splits (both

drawn from SEP-28K and UCLASS), these results represent an optimistic upper bound. The test split also contains the 53-clip Project Boli subset described below.

2. **Cross-corpus held-out (Boli):** 53 clips from the Project Boli dataset, pinned to the test split and therefore entirely unseen during training. This provides an estimate of generalization to new speakers, accents, and recording conditions. Per-class clip counts on Boli are limited (PR: 16, B: 35, SR: 34, WR: 14, IN: 9), so Boli results are noisy but honest.

V. RESULTS AND DISCUSSION

A. In-Distribution Classification Performance

Table 2 presents selected per-class F1 scores (prolongation and block) for all seven arms on the in-distribution test set, with both default (0.5) and tuned thresholds.

Arm	PR (0.5)	PR (t)	BL (0.5)	BL (t)	SR (0.5)	SR (t)	...
arm01_5x_w2v2	.511	.518	.109	.154	—	—	—
arm02_mt_frz3	.490	.522	.113	.160	—	—	—
arm03_mt_frz20	.142	.339	.035	.295	—	—	—
arm04_cnn_pool	.188	.253	.438	.359	—	—	—
arm05_cnn_single	.221	.253	.451	.441	—	—	—
arm06_cnn_lstm	.248	.259	.426	.521	—	—	—
arm07_cnn_tf	.255	.264	.402	.477	—	—	—

Table 2: Selected in-distribution test F1 scores at default (0.5) and tuned thresholds (PR = prolongation, BL = block).

The **multitask Wav2Vec2** with 3-epoch backbone freezing (arm02) achieves the highest overall test macro F1 (0.5215 at tuned thresholds), demonstrating that the shared backbone provides a strong foundation while reducing the effective number of independent models from five to one and using $\approx 4.9\times$ fewer parameters than five separate classifiers. The **five-binary Wav2Vec2 classifier** (arm01) follows closely at 0.5183, benefiting from dedicated per-class optimization without interference between dysfluency types.

The **multitask Wav2Vec2 with 20-epoch freezing** (arm03) dramatically underperforms (test F1 = 0.142 at default, 0.339 at tuned), confirming that excessive backbone freezing prevents the model from learning task-specific representations. This finding has practical implications: backbone freezing is a critical hyperparameter that must be carefully tuned.

CNN-based models (arms04–07) achieve substantially lower in-distribution F1 (0.188–0.255 at default thresholds), reflecting the limited capacity of small convolutional networks (300K–540K parameters) to capture the complex acoustic patterns of dysfluent speech compared to the 94M-parameter Wav2Vec2 backbone.

B. Cross-Corpus Generalization (Boli)

Table 3 presents F1 scores on the Boli cross-corpus held-out set.

Arm	Boli F1 (0.5)	Boli F1 (tuned)
arm01_5x_w2v2	0.109	0.154
arm02_mt_frz3	0.113	0.160
arm03_mt_frz20	0.035	0.295
arm04_cnn_pool	0.438	0.359
arm05_cnn_single	0.451	0.441
arm06_cnn_lstm	0.426	0.521
arm07_cnn_tf	0.402	0.477

Table 3: Cross-corpus F1 on the Boli held-out set (53 clips, 5 Indian languages).

A striking reversal is observed: **CNN-based models significantly outperform Wav2Vec2 models on cross-corpus data**. The CNN-LSTM model (arm06) achieves the highest Boli F1 (0.521 at tuned thresholds) — more than 3× the Wav2Vec2 five-binary classifier (0.154). This can be attributed to two factors:

1. **Parameter efficiency:** CNN models with 300K–540K parameters are less prone to overfitting to speaker-specific characteristics of the training data (SEP-28K/UCLASS), while the 94M-parameter Wav2Vec2 models memorize training-set idiosyncrasies that do not transfer to unseen speakers and accents.
2. **Spectrogram invariance:** Mel-spectrogram representations are inherently more robust to speaker variability than raw-waveform embeddings, as they compress fine-grained speaker identity information while preserving coarse spectral patterns relevant to dysfluency detection.

This finding has important implications for clinical deployment: models that appear superior in-distribution may fail catastrophically on new populations, and CNN-based architectures may be preferable for low-resource or multilingual settings despite lower in-distribution performance.

C. Per-Class Analysis

Detailed per-class F1 scores on the test set (at tuned thresholds) are presented in Table 4.

Class	Precision	Recall	F1	AUROC	Support
Prolongation	0.467	0.500	0.483	0.844	400
Block	0.387	0.380	0.383	0.735	553
Sound Rep.	0.460	0.658	0.541	0.860	494
Word Rep.	0.413	0.492	0.449	0.830	366
Interjection	0.731	0.772	0.751	0.931	834
Macro Avg	0.492	0.560	0.522	0.860	—

Table 4: Per-class classification metrics for the multitask Wav2Vec2 (arm02) on the test set at tuned thresholds.

Several key observations emerge:

1. **Interjection** is the most detectable dysfluency type (F1 = 0.751, AUROC = 0.931). Filler words (“um,” “uh,” “like”) have distinct acoustic signatures that are well-separated from fluent speech, making them relatively easy for the model to identify.
2. **Block** is the most challenging type (F1 = 0.383, AUROC = 0.735). Silent pauses and tense pauses before speech are acoustically similar to normal speech pauses, making them difficult to distinguish without contextual information.
3. **Sound repetition** achieves the second-highest F1 (0.541) with the highest recall (0.658), indicating the model is effective at catching sound repetitions but with lower precision.
4. The large gap between AUROC and F1 across all classes (e.g., prolongation AUROC = 0.844 vs. F1 = 0.483) indicates threshold miscalibration rather than poor ranking ability — the model’s probability outputs are informative but not well-calibrated.

D. Localization Performance

Table 5 compares localization metrics for the two localizers — the CNN spectrogram localizer and the Wav2Vec2 temporal localizer — on the test and Boli held-out sets.

Metric	CNN Test	CNN Boli	W2V2 Test	W2V2 Boli
Frame Precision	0.485	0.154	0.676	0.177
Frame Recall	0.690	0.918	0.065	0.023
Frame F1	0.570	0.264	0.119	0.040
Frame Specificity	0.373	0.284	0.973	0.985
Detection Accuracy	0.222	0.375	0.210	0.000
Mean IoU	0.707	0.614	0.751	0.000
False Alarms/min	55.89	33.86	8.95	11.16

Table 5: Localization metrics for the CNN spectrogram localizer and the Wav2Vec2 temporal localizer on test and Boli held-out sets.

The two localizers exhibit complementary operating points. The **Wav2Vec2 temporal localizer** is highly conservative: on the test set it achieves high frame precision (0.676) but very low recall (0.065), missing most dysfluency events while producing relatively few false alarms (8.95/min); when it does detect an event, temporal accuracy is reasonable (mean IoU = 0.751). On Boli it detects no events at all. The **CNN spectrogram localizer** takes the opposite stance: substantially higher recall (0.690 test, 0.918 Boli) and higher frame F1 (0.570 test, 0.264 Boli), but at the cost of a high false alarm rate (55.9/min test, 33.9/min Boli) and low specificity. Neither localizer is yet suitable for standalone clinical deployment: the Wav2Vec2 variant misses too many events, while the CNN variant flags far too many. The registry’s combiner mitigates this by fusing localizer regions with the classifier’s per-frame saliency maps, so that conservative temporal anchors are complemented by class-aware evidence; improving the precision–recall balance of both localizers is a priority for future work.

E. Computational Considerations

The comparison between Wav2Vec2 and CNN models highlights a fundamental trade-off:

Model Family	Params	Test F1	Boli F1
Wav2Vec2 (5× binary)	94.6M	0.518	0.154
Wav2Vec2 (multitask)	97.3M	0.522	0.160
CNN (best)	540K	0.264	0.521

Table 6: Trade-off between parameter count and generalization performance.

Wav2Vec2 models achieve 2× higher in-distribution F1 but use 180× more parameters and exhibit 3.4× worse cross-corpus performance. For resource-constrained or multilingual deployment, CNN models offer a compelling alternative with competitive cross-corpus performance and negligible computational overhead.

F. Comparison with Published Results

Our multitask Wav2Vec2 results (macro F1 = 0.522 at tuned thresholds) are consistent with published benchmarks: Bayerl et al. [1] reported macro F1 in the range of 0.56 – 0.63 on FluencyBank, and Miyahara et al. [5] reported per-class F1 ranging from 0.30 (block) to 0.78 (interjection) on SEP-28K. Our interjection F1 (0.751) closely matches their 0.78, and our block F1 (0.383) exceeds their 0.30, suggesting our shared-backbone architecture provides modest improvement for the most challenging class. The macro-averaged F1 of 0.522 falls within the literature-expected range of 0.45 – 0.65 for Wav2Vec2-based classifiers on this task.

G. Limitations

1. **Same-speaker overlap:** The test set contains speakers from the same source datasets as training (SEP-28K, UCLASS), inflating in-distribution metrics. Boli provides a more honest cross-corpus evaluation but is limited in size (53 clips).
2. **Single seed:** All results use seed = 42. A multi-seed evaluation would provide confidence intervals and more robust comparisons.

3. **Localization recall:** The Wav2Vec2 localizer’s low recall (0.065) and the CNN localizer’s high false-alarm rate (55.9/min on test) limit clinical utility. Improving the precision–recall balance of both localizers is a priority for future work.
4. **Binary thresholds:** Per-class thresholds are tuned on the validation set only, which may not be optimal for all deployment scenarios.

H. Practical Implications

For clinical stutter detection systems, our results suggest:

1. **Use Wav2Vec2 for high-accuracy screening** where in-distribution performance is the priority (e.g., monitoring known patient populations).
2. **Use CNN models for generalization-critical deployment** where unseen speakers, accents, or recording conditions are expected (e.g., population-level screening tools).
3. **Always report cross-corpus metrics** alongside in-distribution metrics to provide honest performance estimates.
4. **Interjection detection is reliable** and can serve as an anchor for user trust; block detection requires additional complementary signals (e.g., silence detection heuristics).

VI. CONCLUSION

This paper presented Swaraaha, an end-to-end speech dysfluency classification and localization system, and provided a systematic comparative study of seven architectural variants across three model families. Our key findings are:

1. The **shared-backbone multitask classifier** achieves the highest in-distribution performance ($F1 = 0.5215$), while the **five-binary variant** follows closely ($F1 = 0.5183$) using $\approx 4.9\times$ more parameters in total.
2. **CNN-based models** dramatically outperform Wav2Vec2 on cross-corpus evaluation (Boli $F1 = 0.521$ vs. 0.160), revealing a fundamental trade-off between in-distribution accuracy and generalization.
3. **Interjection** is the most reliably detected dysfluency type ($F1 = 0.751$), while **block** remains the most challenging ($F1 = 0.383$), consistent with prior literature.
4. **Backbone freezing duration** is a critical hyperparameter — excessive freezing (20 epochs) degrades tuned macro $F1$ by $\approx 35\%$ compared to moderate freezing (3 epochs).

Future work will focus on: (1) improving localization recall through semi-supervised training with pseudo-labeled SEP-28K data, (2) backbone ablation across XLS-R-300M and HuBERT for multilingual support, (3) multi-seed evaluation with confidence intervals, and (4) integration of stutter-aware ASR for freeform speech analysis.

Swaraaha is open-source and available at <https://github.com/chethanbhat7/Swaraaha>.

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